

COMPASS_Synoptic_LE_Soil-Fluxes_2024_CO2

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0.1 Goals of this code:

1. Reads in raw smart chamber
2. If no metadatafile, write one out & fields will be: Label the IDs pulled from the label, Plot *fixed so that if there are any issues in the label they are good here Drop column – false as long as label looks right adjust time – would be initially populated by the start time in the OR blank adjust observation length – initially populated by the obs length OR blank notes / comment
3. Code reads in the meta data and applies it to the data
 - a. i.e. drops fluxes that have the drop column TRUE or FLAG this
 - b. adjusting times as needed
4. Now you compute fluxes
5. Visualize and look for issues
6. If found, report to meta data file then go back to 3 and reanalyze
7. Apply algorithmic qaqc and generate flags

0.2 Set Up & Load Packages

0.3 Pull in Smart Chamber Files from the Raw Data folder

0.4 label format

0.5 Create a unique Plot_ID, check if there is a meta-data file, if not, create a meta-data file

Warning in unzip(zip_file, files = csv_inside_zip, exdir = "."): error 1 in extracting from zip file

```
`summarise()` has regrouped the output.
i Summaries were computed grouped by InstrumentSerialNumber and Plot_ID.
i Output is grouped by InstrumentSerialNumber.
i Use `summarise(.groups = "drop_last")` to silence this message.
i Use `summarise(.by = c(InstrumentSerialNumber, Plot_ID))` for per-operation
  grouping (`?dplyr::dplyr_by`) instead.
```

```
[1] "CSV file does not exist. It has been created."
```

0.6 Mark fluxes that were retaken - this is the first issue! Need to get all but last flux to be TRUE for drop - fieldnotes with explanation for flux retaken - usually weird zeros, Steven probe not inserted in the soil, or label mistake

```
Warning: There were 6 warnings in `mutate()`.
The first warning was:
i In argument: `drop = letter_order < max(letter_order, na.rm = TRUE)`.
i In group 8: `core_id = "2024_02_02_CRC_UP_25"`.
Caused by warning in `max()`:
! no non-missing arguments to max; returning -Inf
i Run `dplyr::last_dplyr_warnings()` to see the 5 remaining warnings.
```

##Dont need to pull L1 temp data or any other temps like we do for macrophyte because the smartchamber records the temp

0.7 Match up the metadata to the rest of the data

0.8 fixing some Deadbands

##Extracting initial CO2 concentration

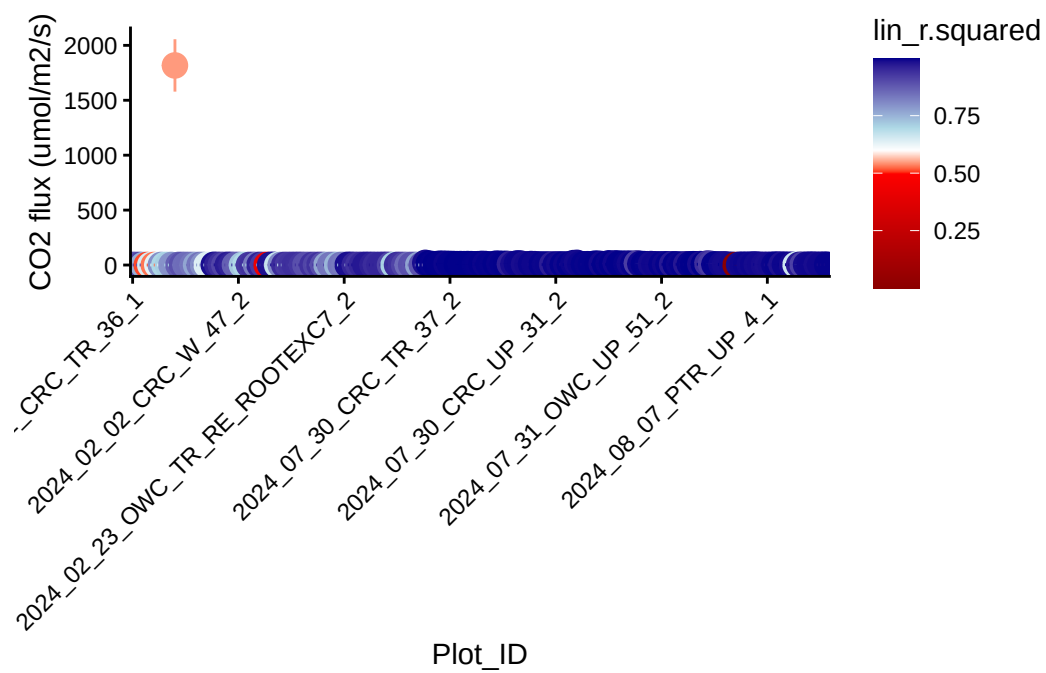
0.9 Change the units from ppb to nmol

Assuming atm = 101325 Pa

Using $R = 8.31446261815324 \text{ m}^3 \text{ Pa K}^{-1} \text{ mol}^{-1}$

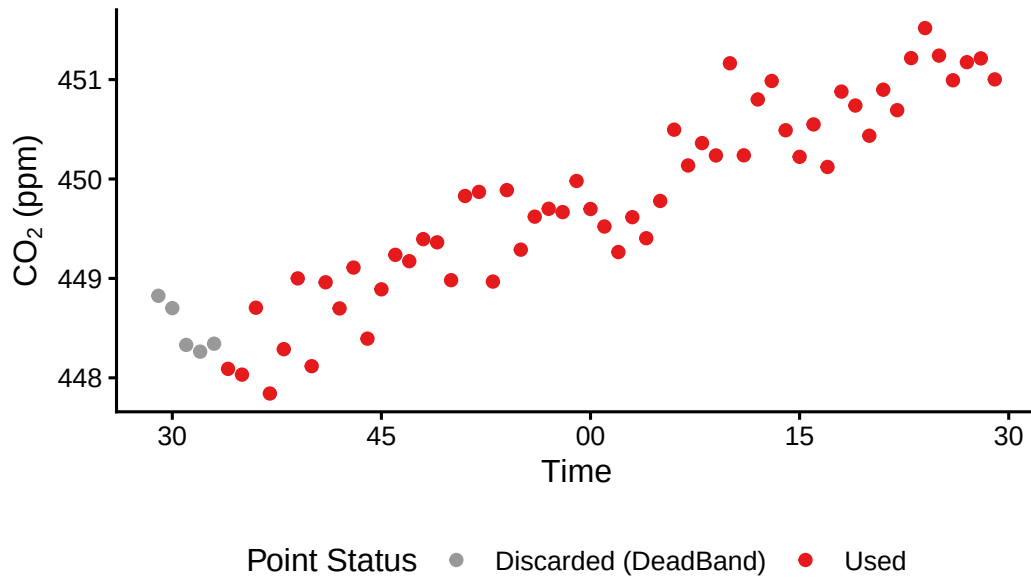
0.10 Calculate fluxes

0.11 Visualize fluxes and sort by r2

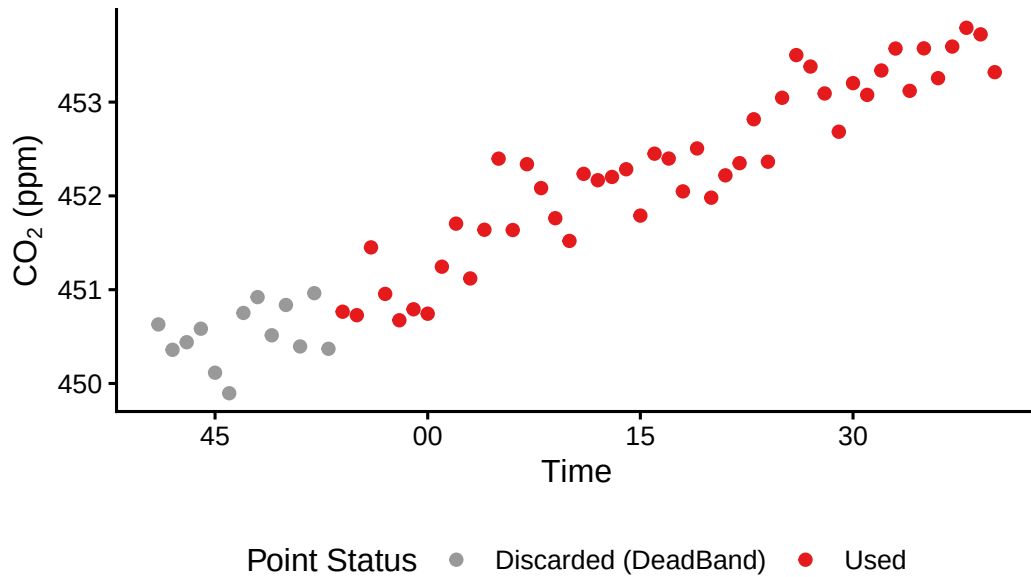


0.12 check the fluxes for which I changed the start time due to lags or ebullition (recorded in notes of metadata)

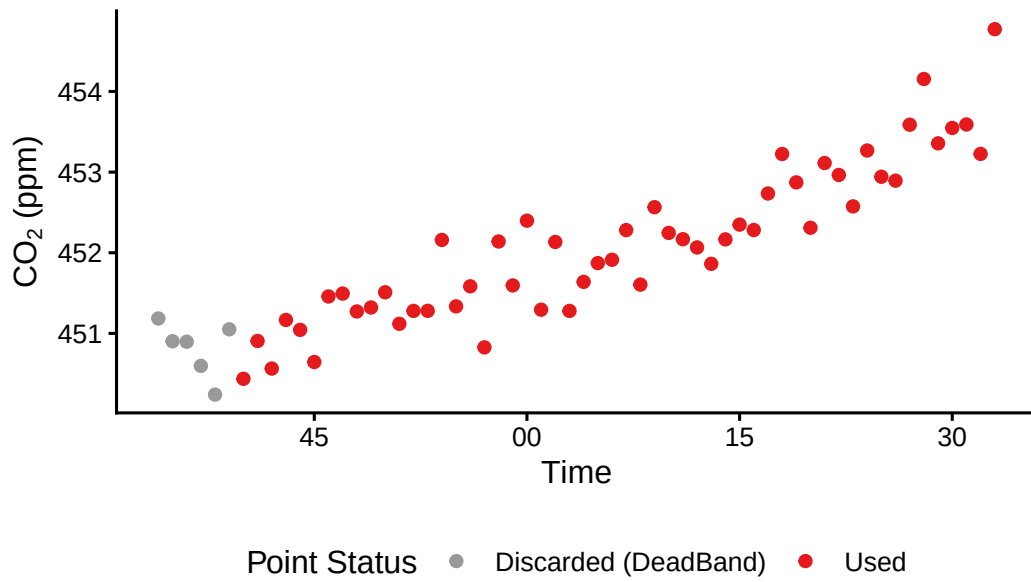
Incubation Period Adjustments: 2024_02_23_OWC_V



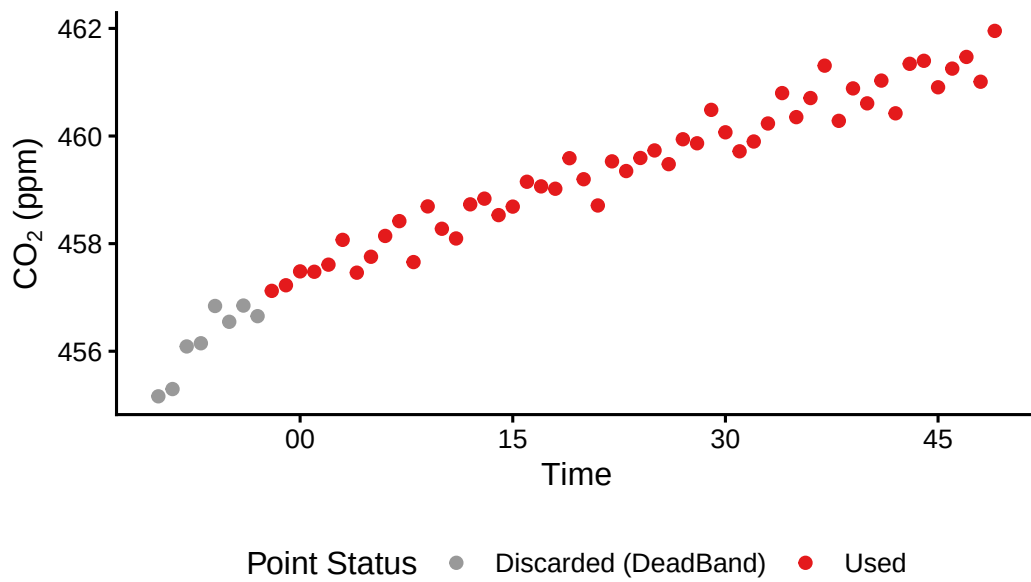
Incubation Period Adjustments: 2024_02_23_OWC_V



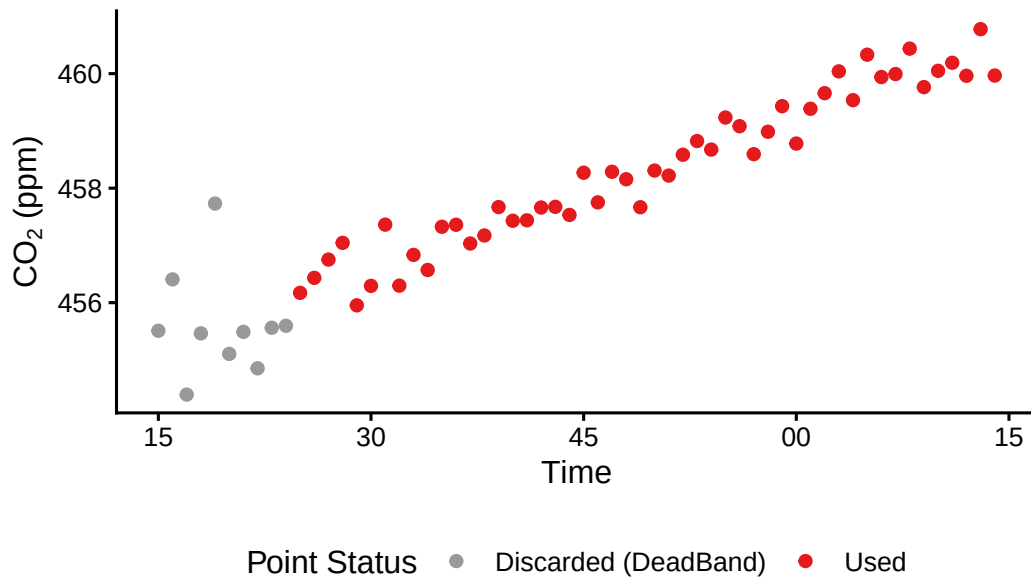
Incubation Period Adjustments: 2024_02_23_OWC_V



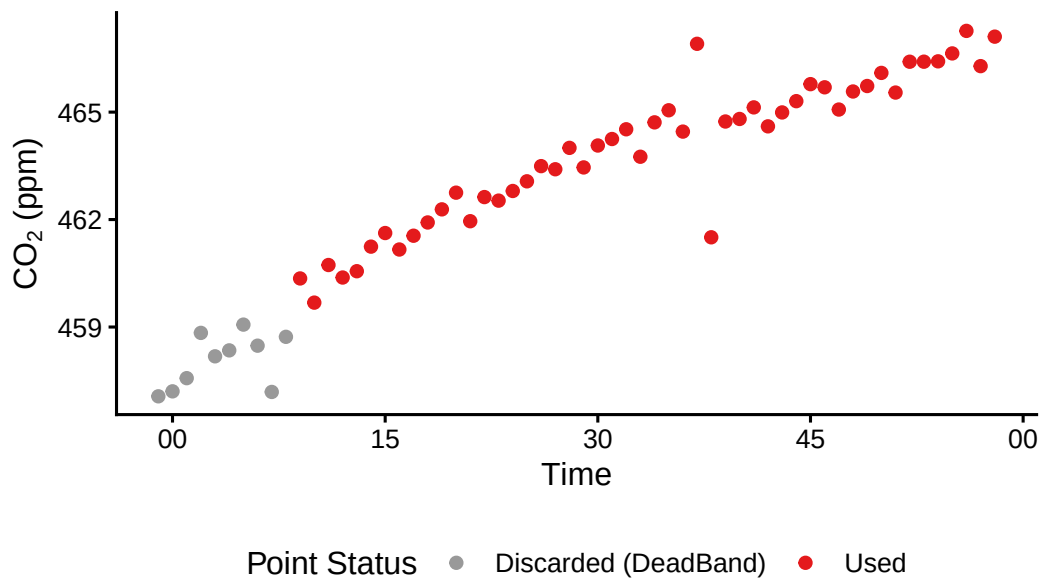
Incubation Period Adjustments: 2024_02_23_OWC_T



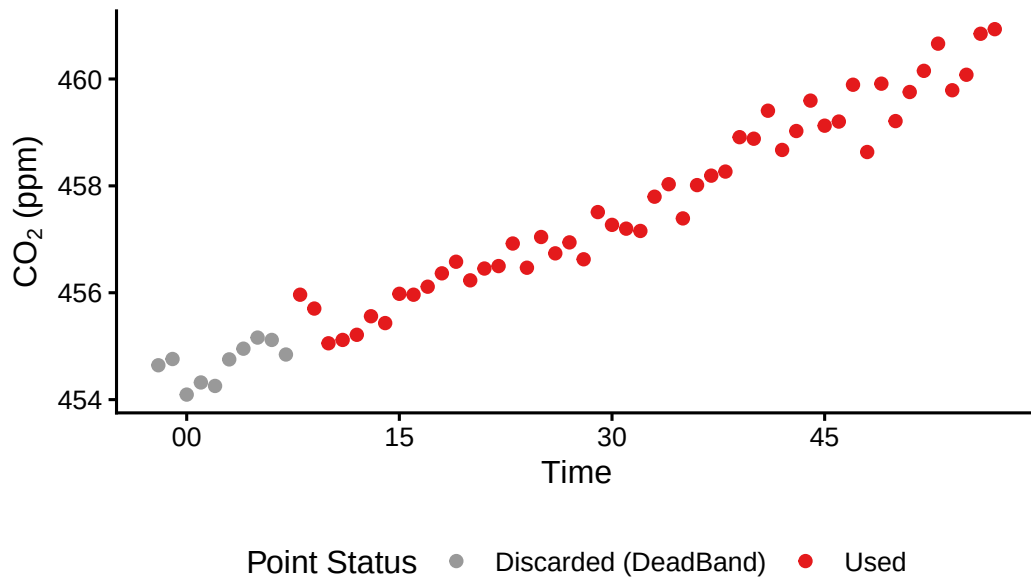
Incubation Period Adjustments: 2024_02_23_OWC_T



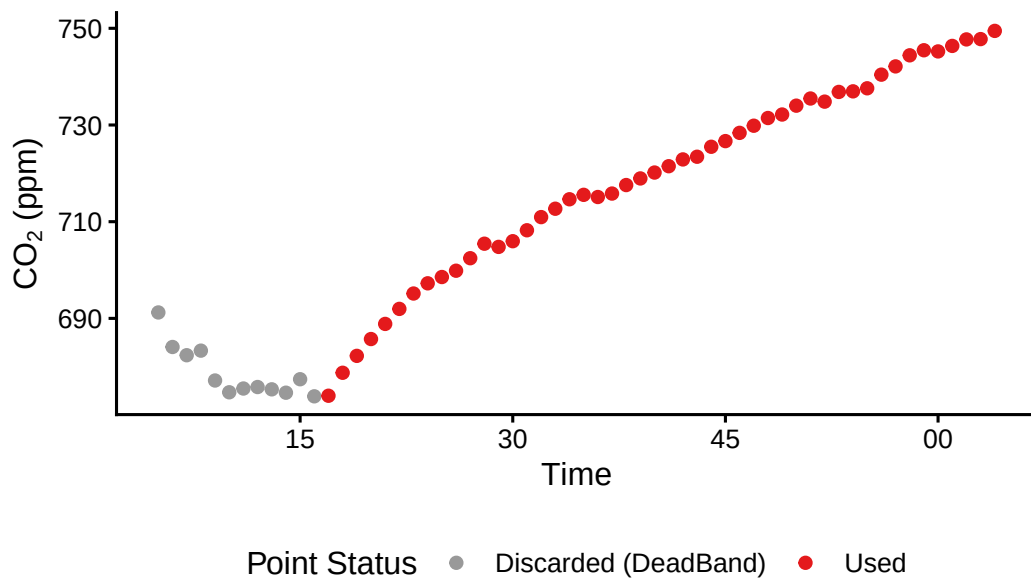
Incubation Period Adjustments: 2024_02_23_OWC_T



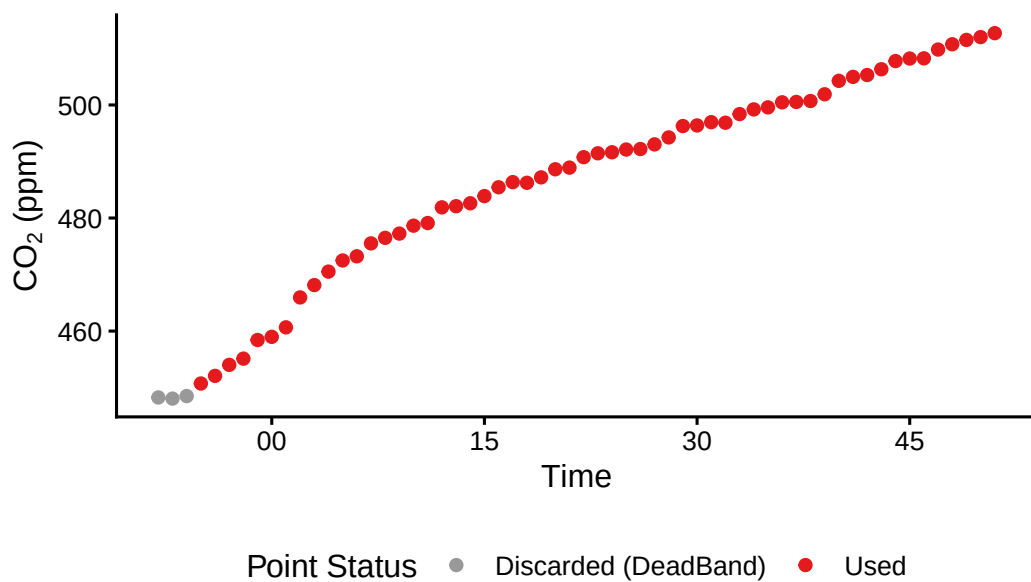
Incubation Period Adjustments: 2024_02_23_OWC_U



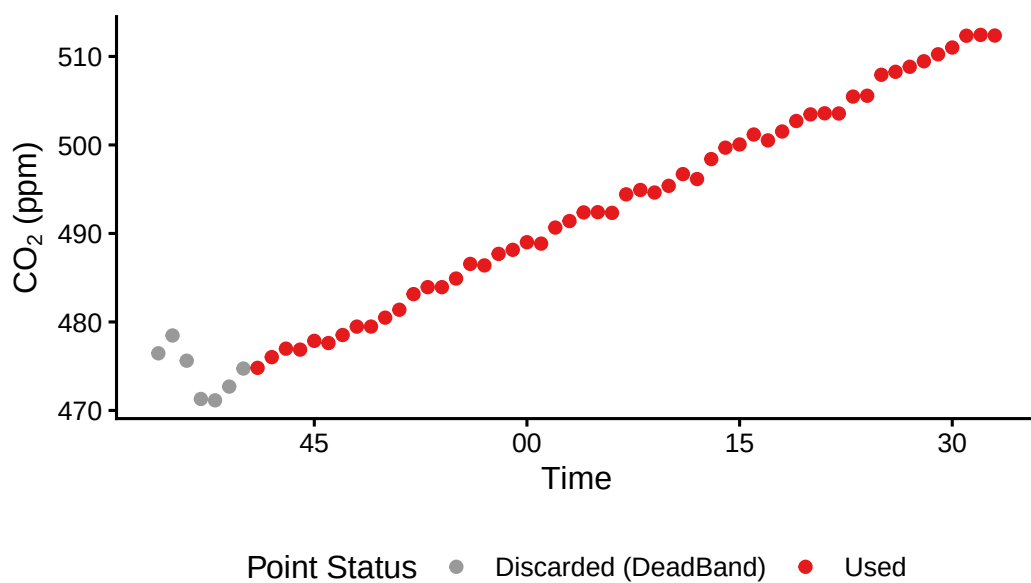
Incubation Period Adjustments: 2024_07_31_OWC_T



Incubation Period Adjustments: 2024_07_31_OWC_U



Incubation Period Adjustments: 2024_07_31_OWC_U

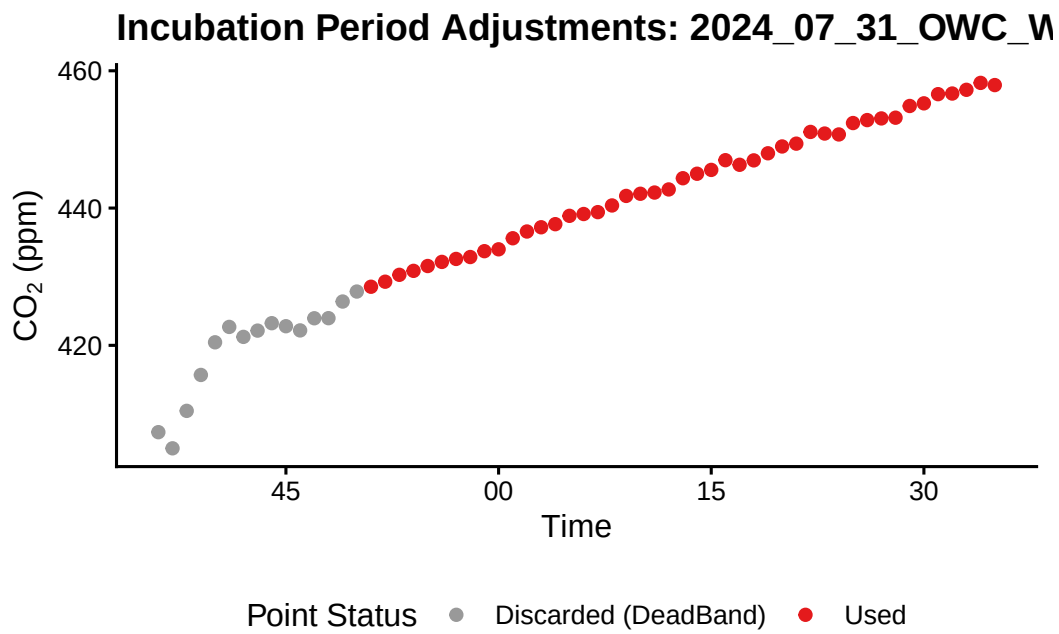
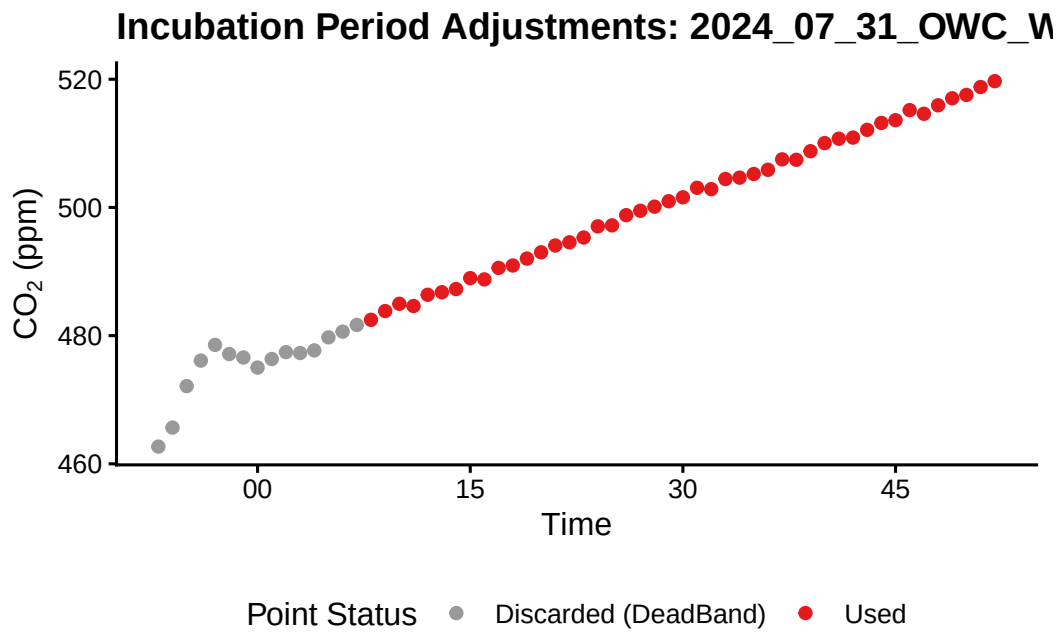


Scatter plot showing CO₂ concentration (ppm) versus Time (h) for the first 16 hours. The concentration decreases from approximately 467 ppm at 0h to a minimum of about 459 ppm at 25h, then increases steadily to about 495 ppm at 16h. The data points are colored grey for the first 30 hours and red for the subsequent 16 hours.

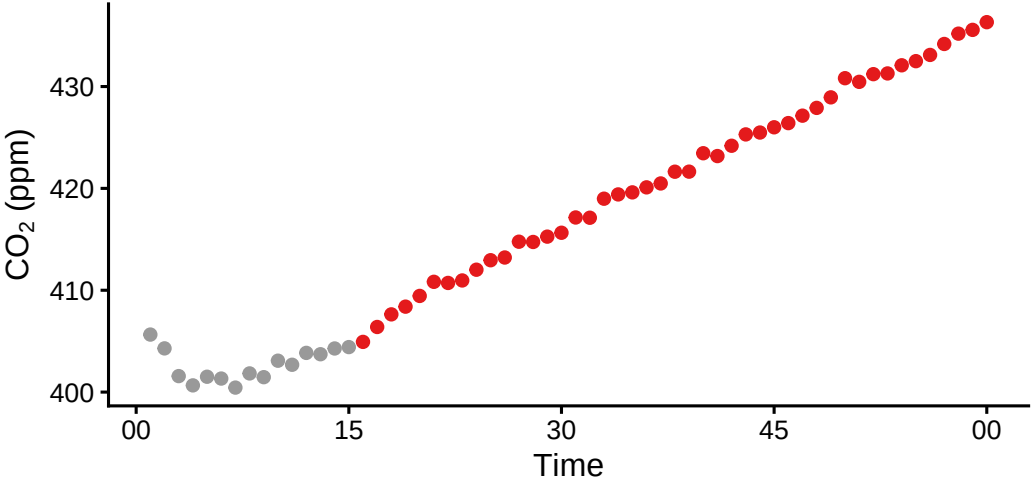
The scatter plot displays the change in CO₂ concentration over time. The y-axis represents CO₂ (ppm) from 475 to 575, and the x-axis represents Time (min) from 25 to 45. The data points show a characteristic U-shaped trend, with a minimum concentration of approximately 485 ppm occurring around 35 minutes. The initial decrease is marked with red dots, and the subsequent increase is marked with blue dots.

Time (min)	CO ₂ (ppm)
29	500
30	490
31	488
32	485
33	488
34	490
35	495
36	500
37	505
38	508
39	512
40	518
41	525
42	530
43	535
44	538
45	540
46	545
47	548
48	550
49	552
50	555
51	558
52	560
53	562
54	565
55	568
56	570
57	572
58	575
59	578
60	580

10

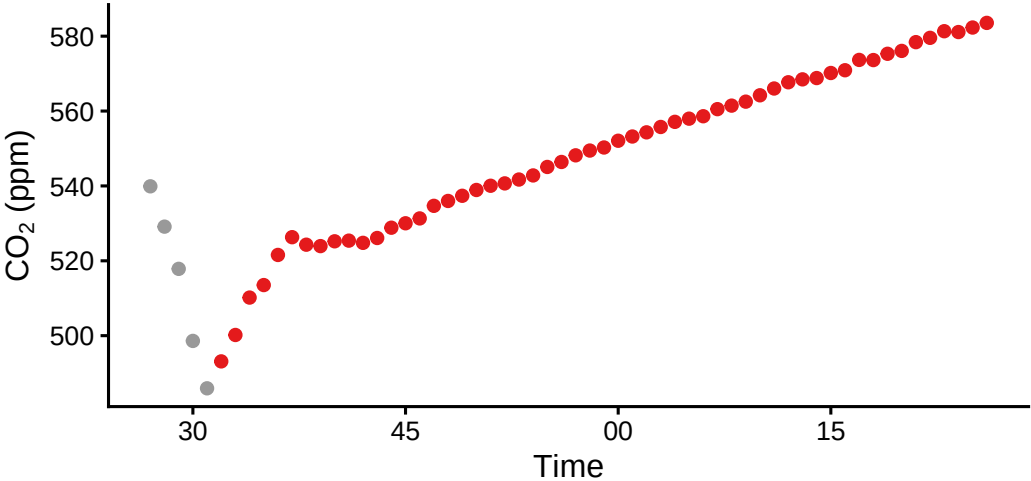


Incubation Period Adjustments: 2024_07_31_OWC_W

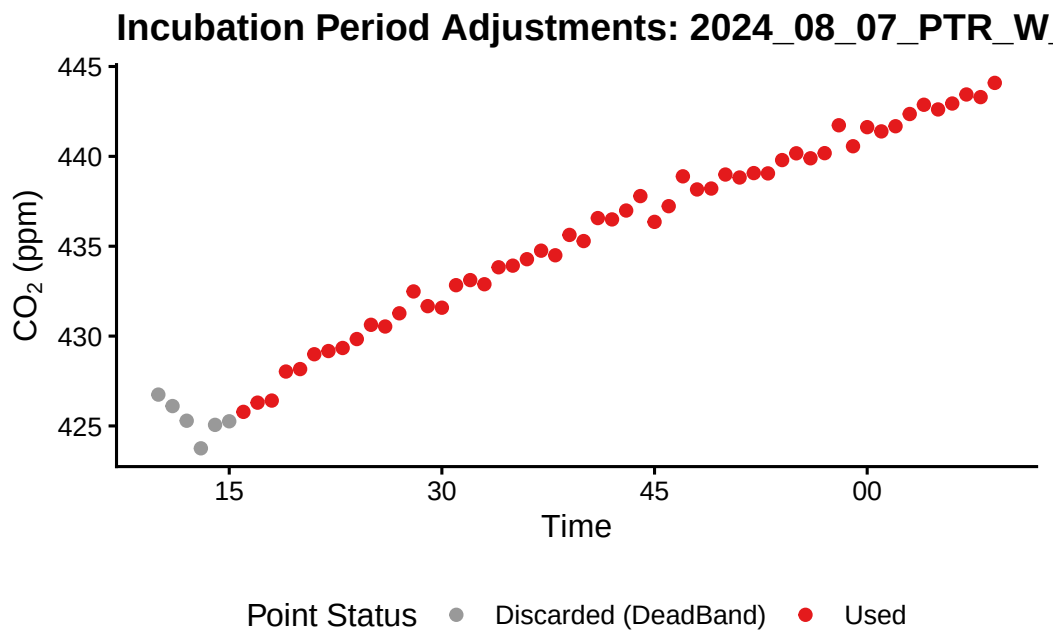
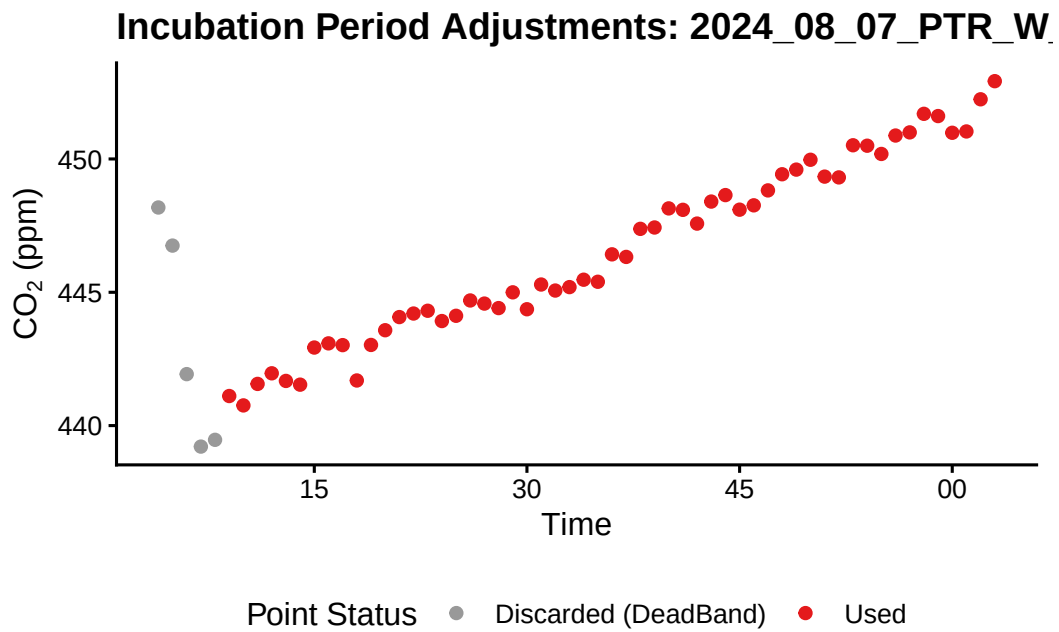


Point Status ● Discarded (DeadBand) ● Used

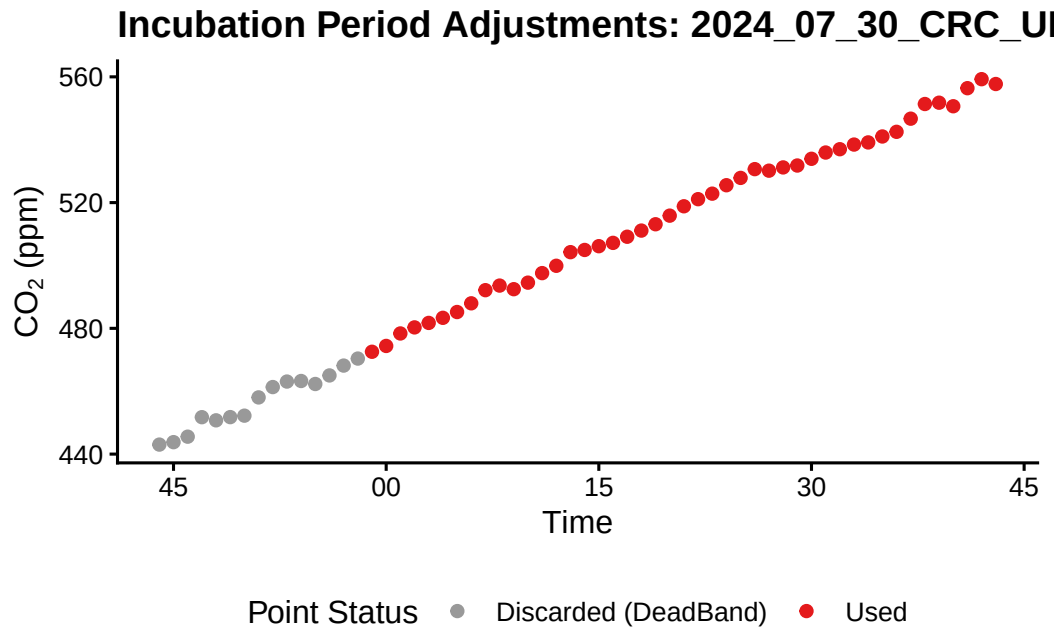
Incubation Period Adjustments: 2024_07_31_OWC_T



Point Status ● Discarded (DeadBand) ● Used

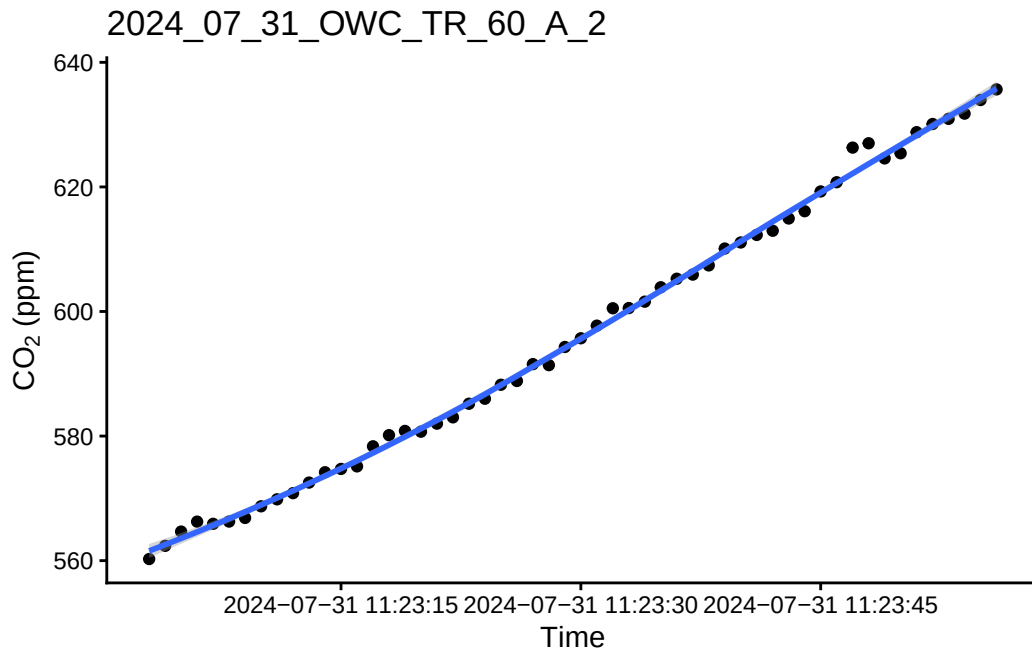




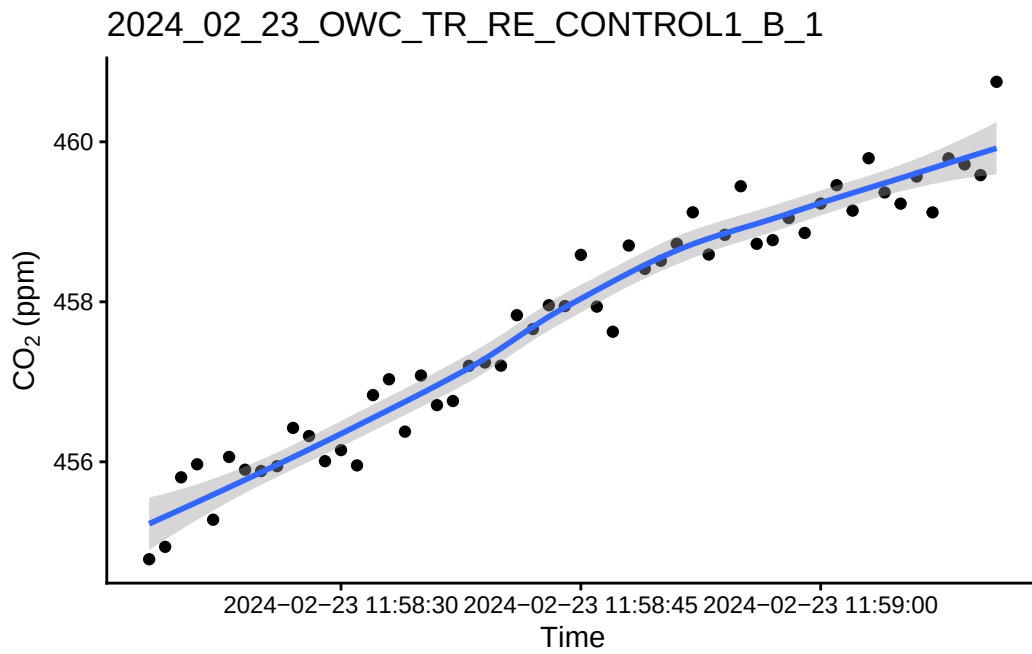


0.13 Spot check a few fluxes with $R^2 > 0.9$

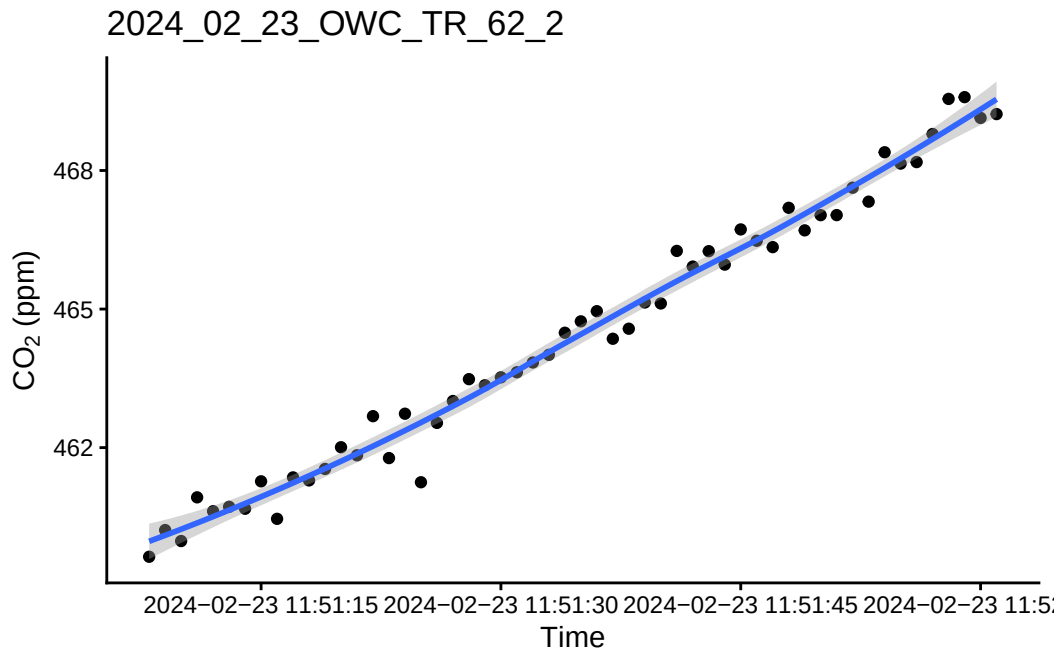
``geom_smooth()`` using method = 'loess' and formula = 'y ~ x'



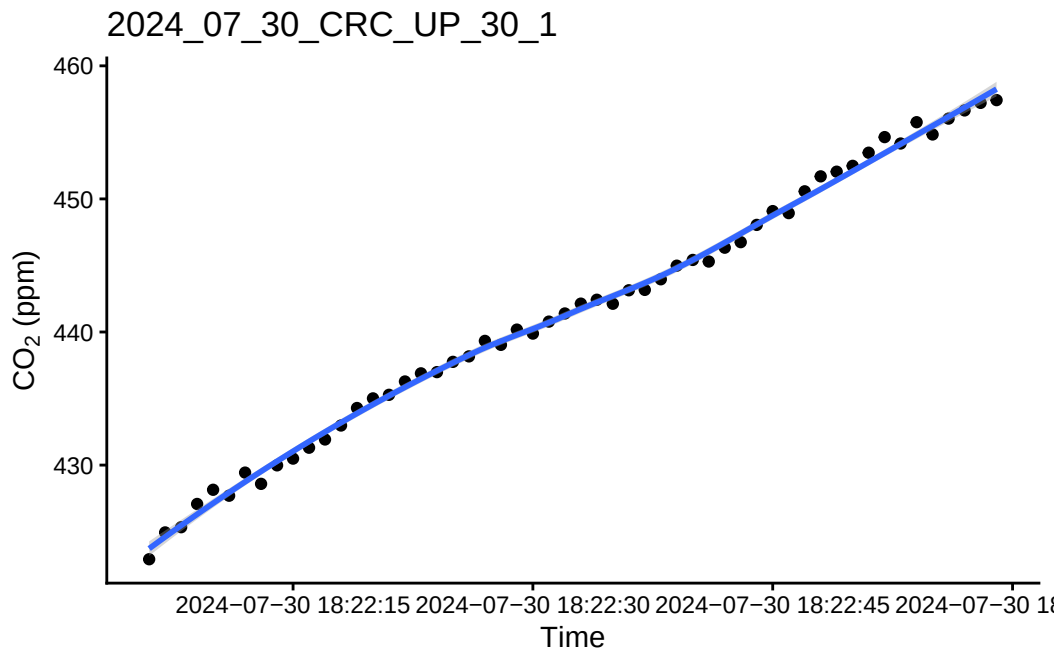
``geom_smooth()`` using method = 'loess' and formula = 'y ~ x'



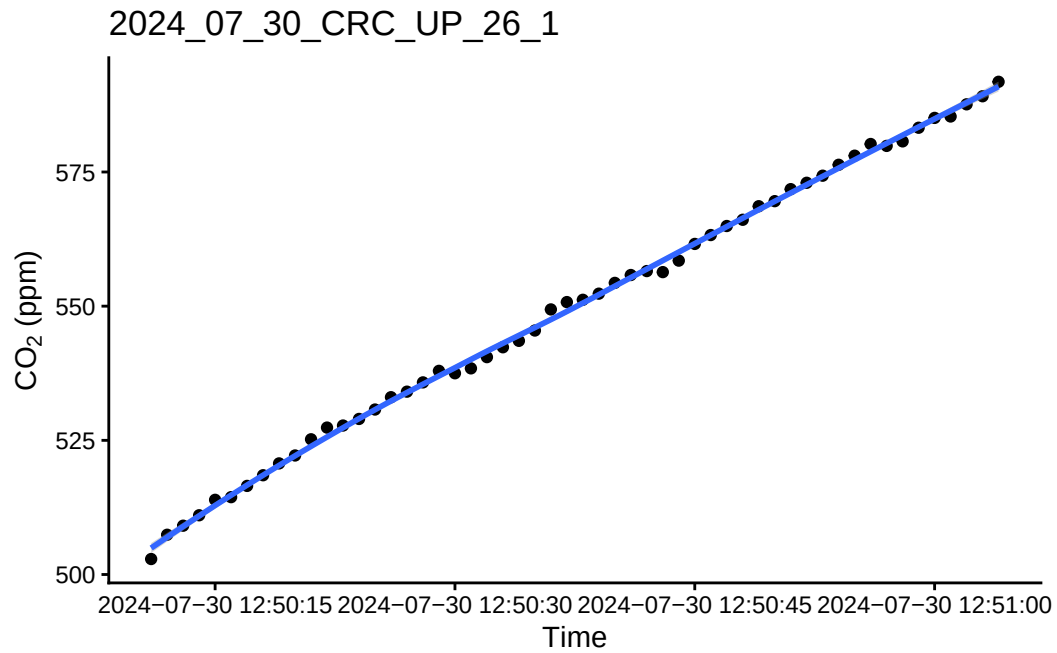
``geom_smooth()`` using method = 'loess' and formula = 'y ~ x'



``geom_smooth()`` using method = 'loess' and formula = 'y ~ x'

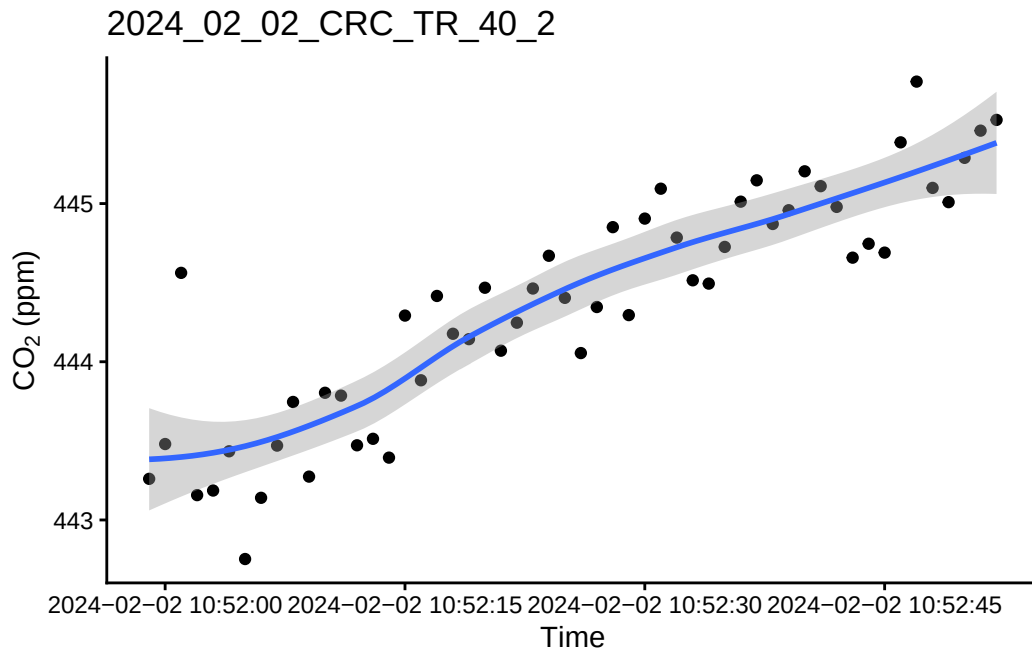


``geom_smooth()`` using method = 'loess' and formula = 'y ~ x'

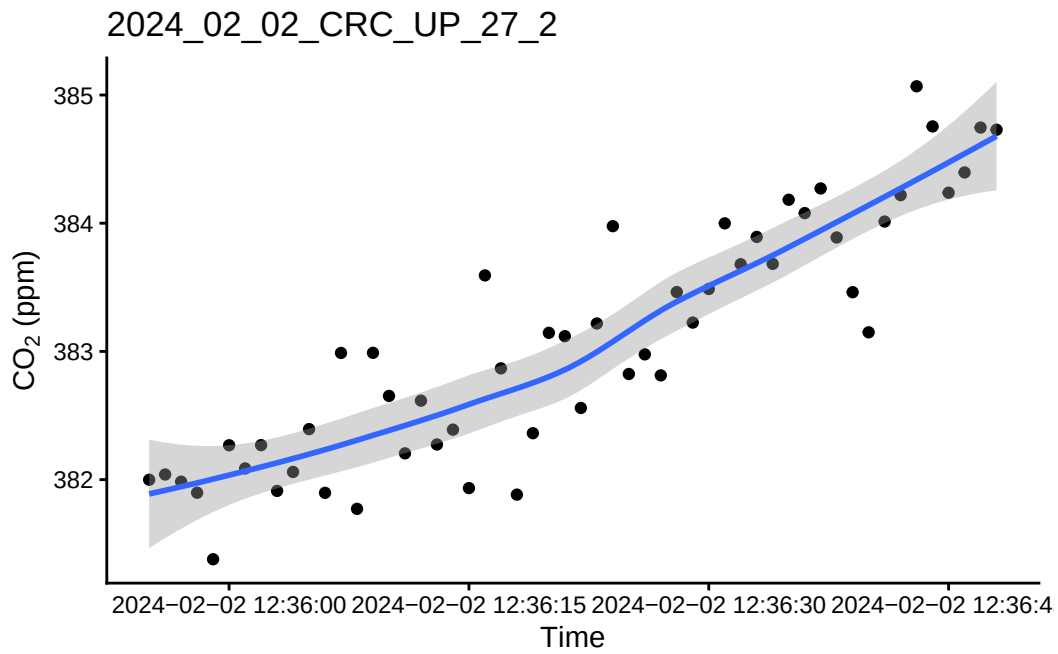


0.14 Take a look at the fluxes <0.9 and a flux estimate below 1

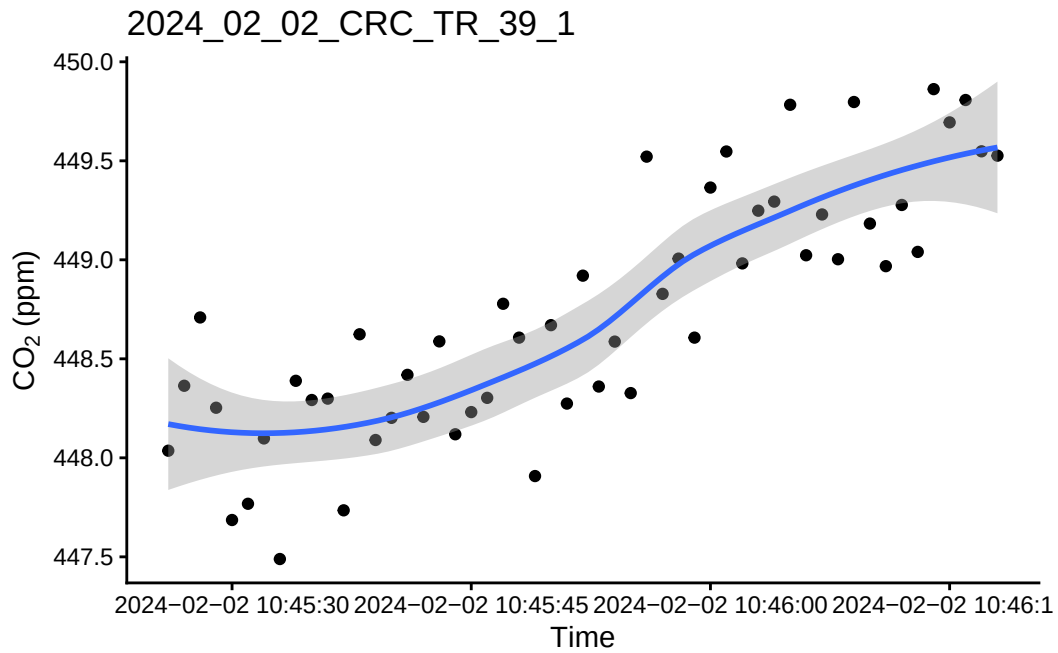
``geom_smooth()`` using `method = 'loess'` and `formula = 'y ~ x'`



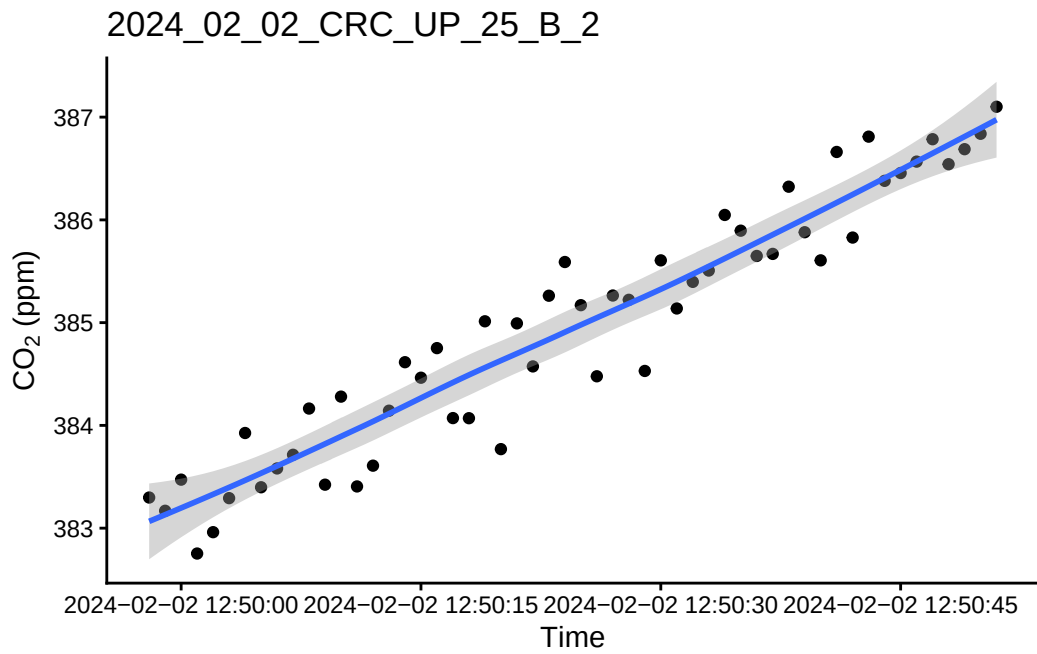
``geom_smooth()`` using method = 'loess' and formula = 'y ~ x'



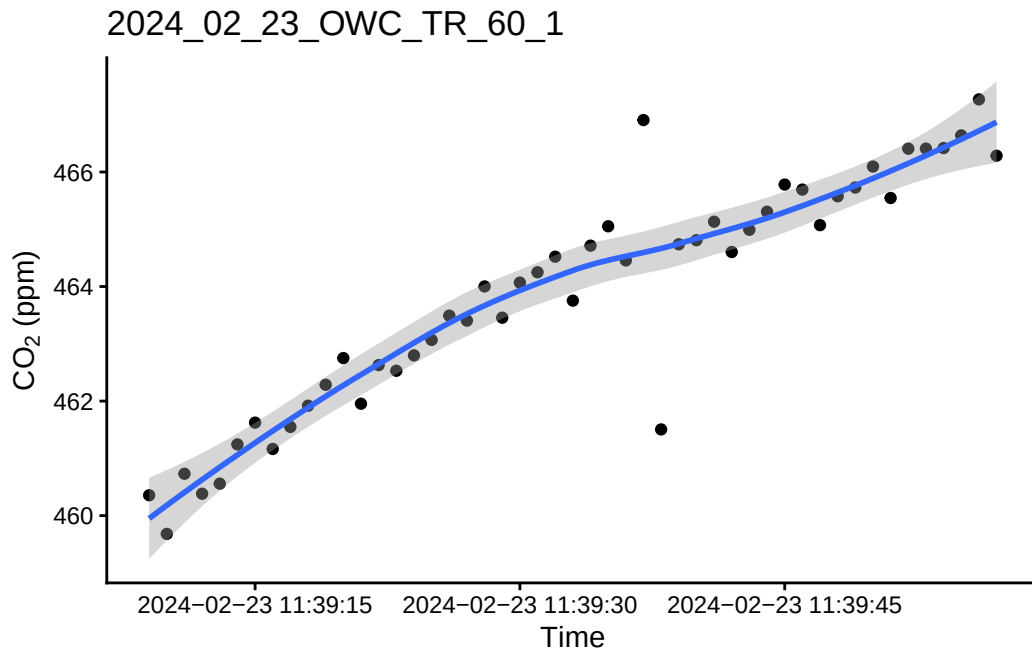
``geom_smooth()`` using method = 'loess' and formula = 'y ~ x'



``geom_smooth()`` using method = 'loess' and formula = 'y ~ x'



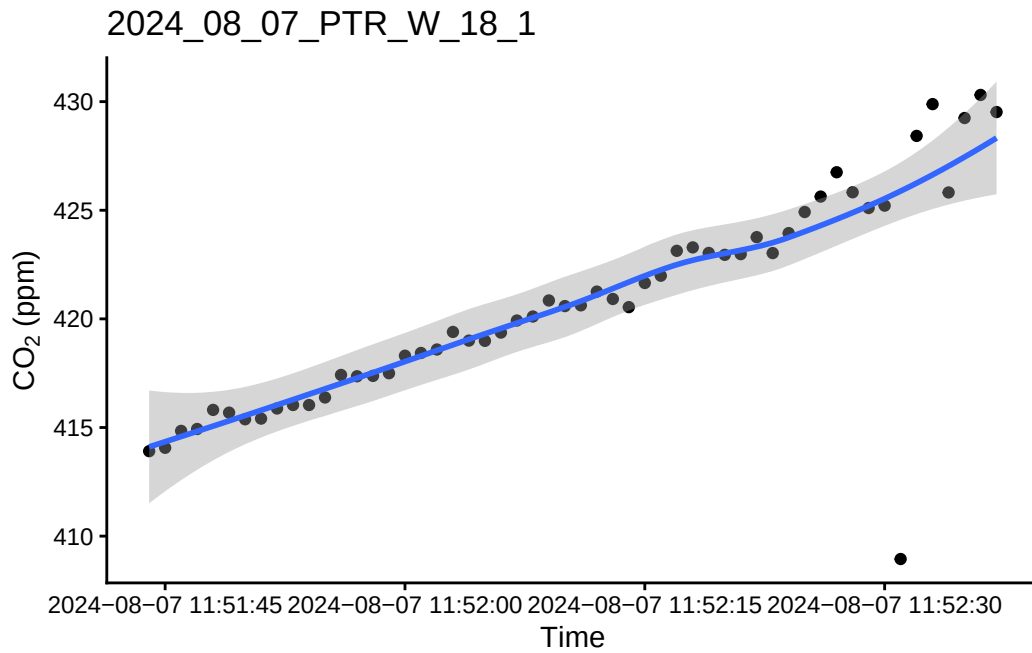
``geom_smooth()`` using method = 'loess' and formula = 'y ~ x'



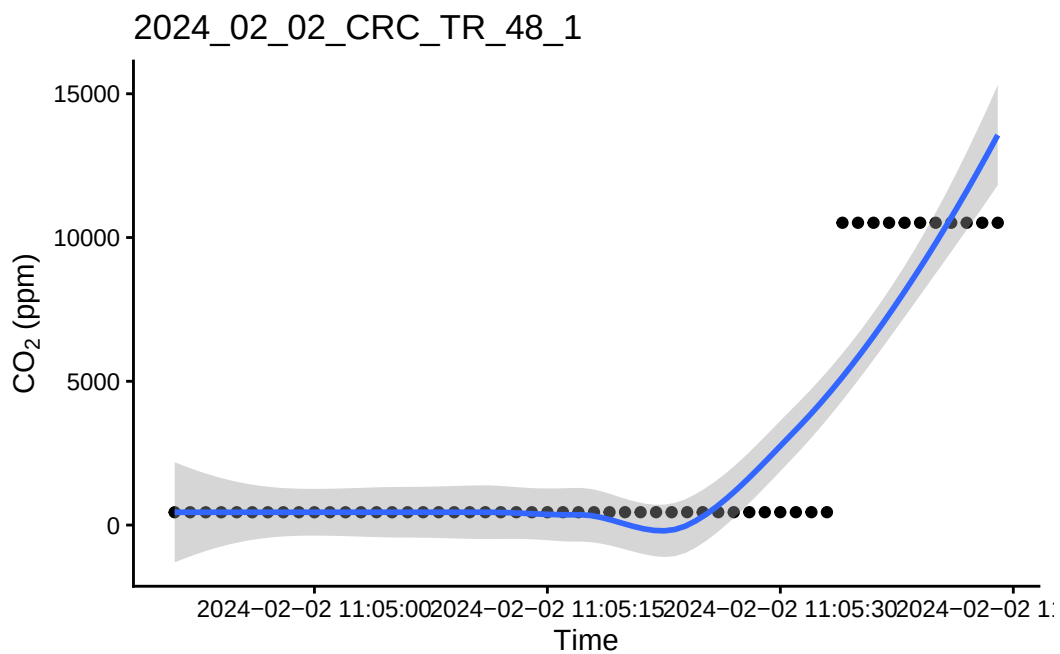
0.15 Take a look at the bad fluxes <0.9 and have a high flux estimate

1 See if we want to remove these fluxes or if we want to try and recalculate these ones?

```
`geom_smooth()` using method = 'loess' and formula = 'y ~ x'
```

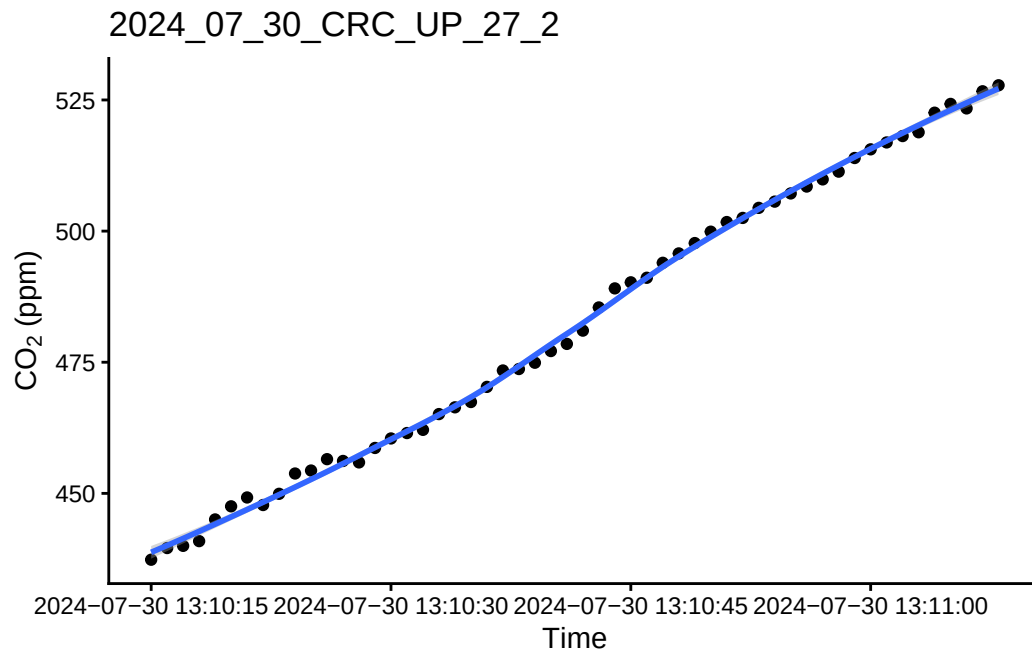


``geom_smooth()`` using method = 'loess' and formula = 'y ~ x'

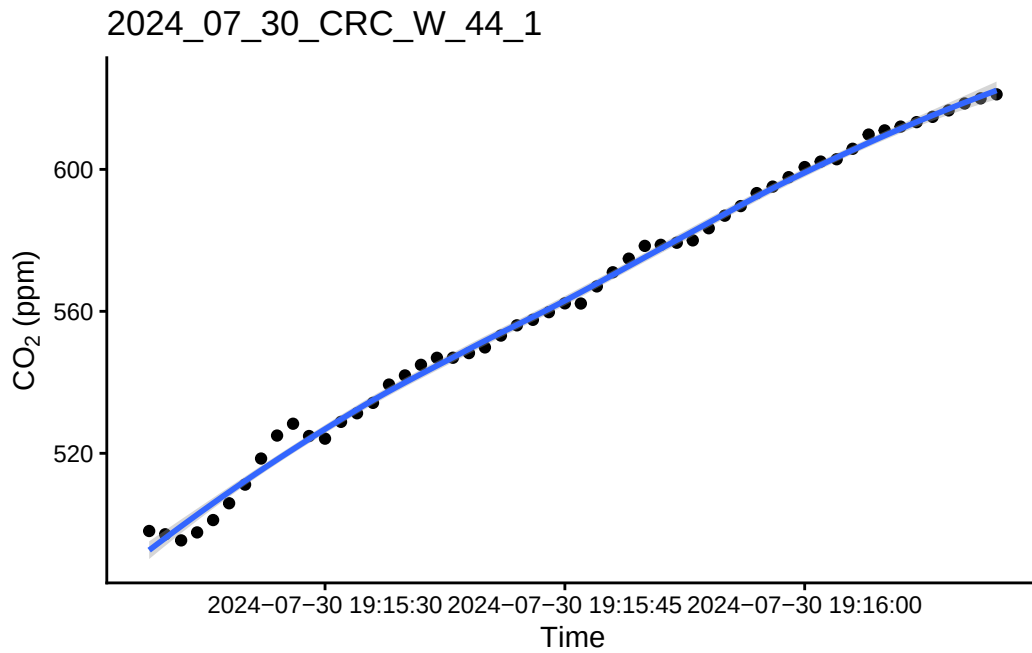


1.1 Look at the really high or really low ones to see if there is a timing issue or something

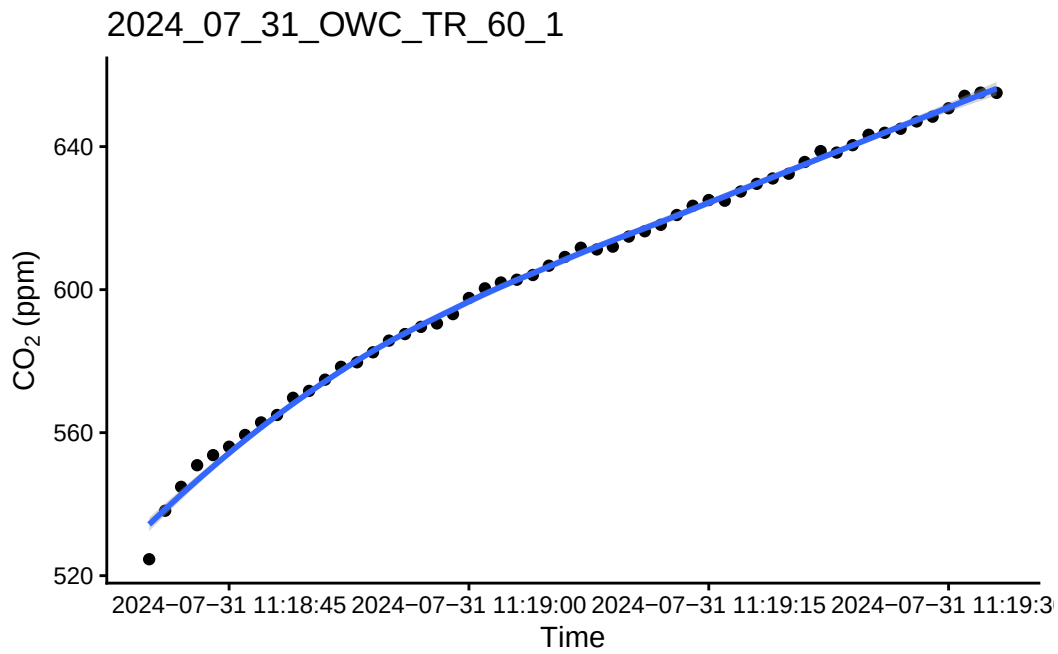
```
`geom_smooth()` using method = 'loess' and formula = 'y ~ x'
```



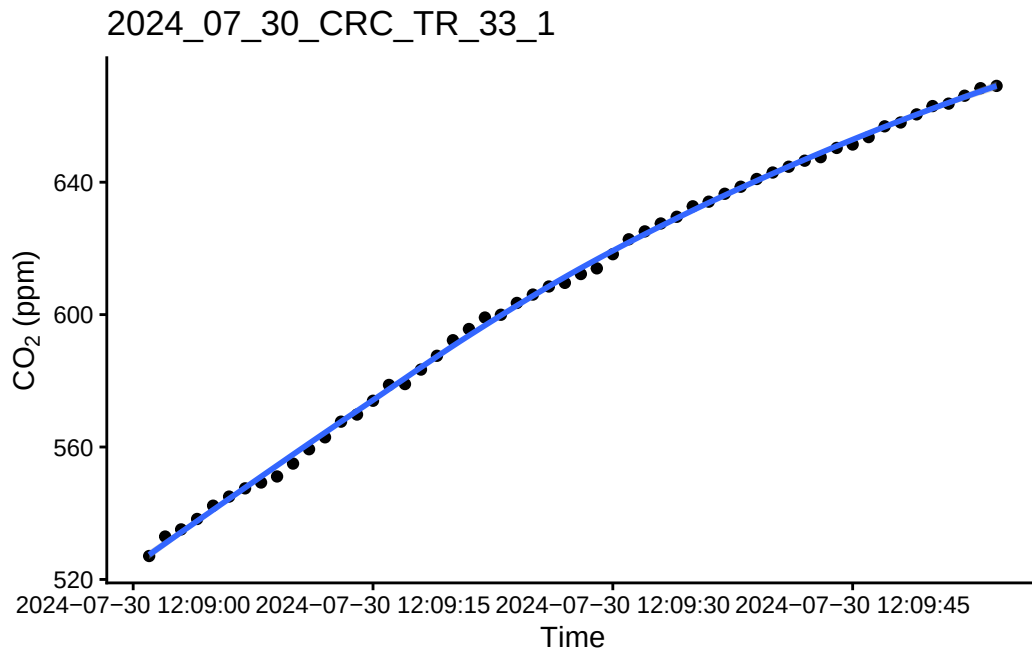
```
`geom_smooth()` using method = 'loess' and formula = 'y ~ x'
```



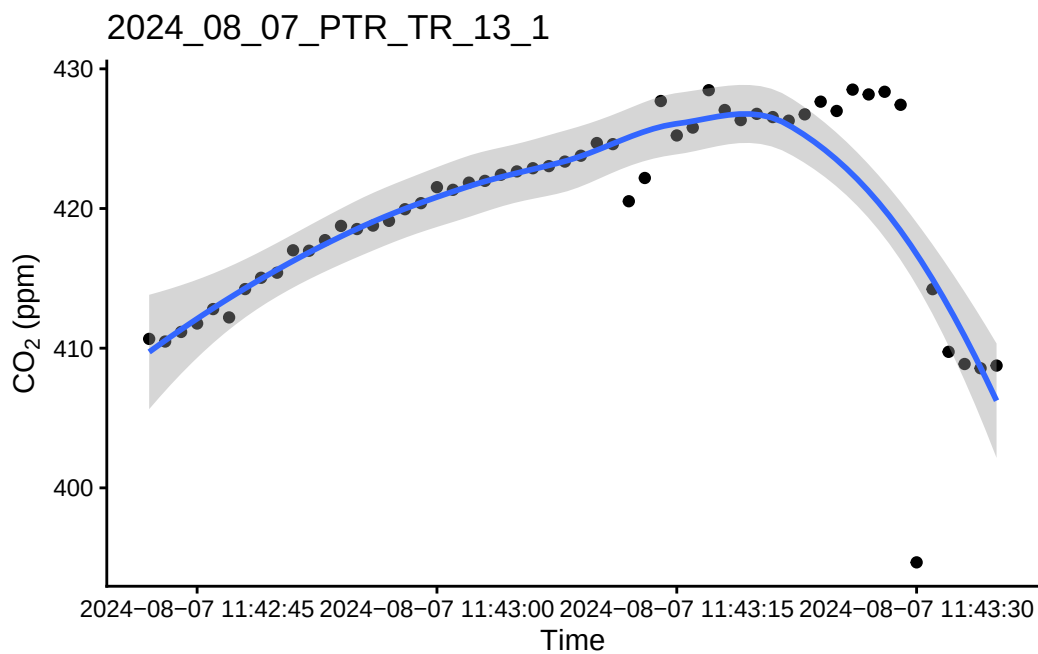
``geom_smooth()`` using method = 'loess' and formula = 'y ~ x'



``geom_smooth()`` using method = 'loess' and formula = 'y ~ x'

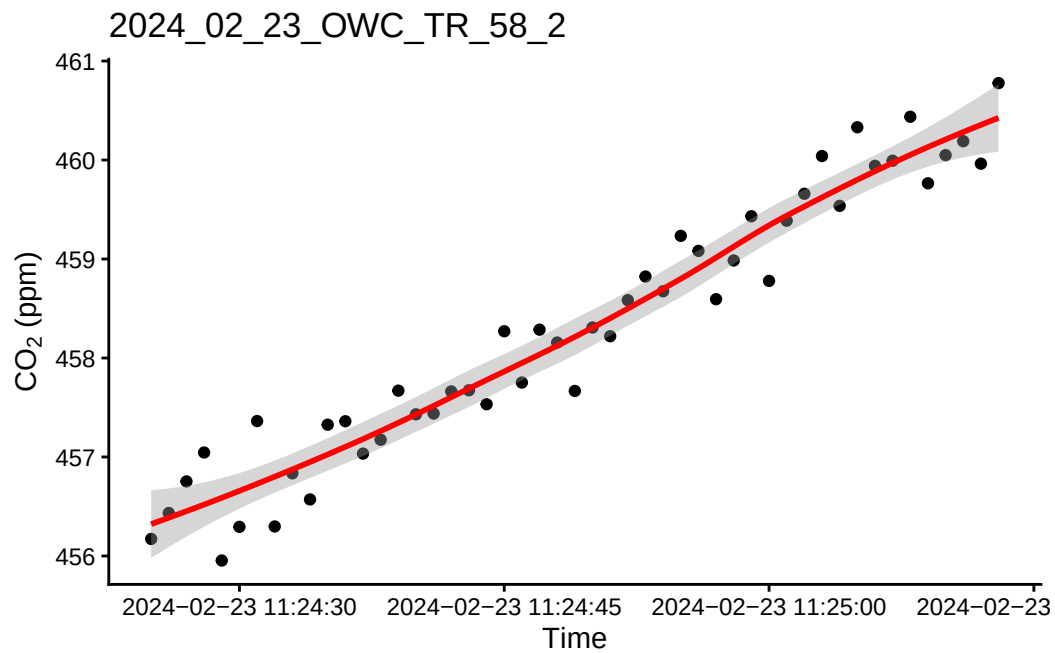


``geom_smooth()`` using method = 'loess' and formula = 'y ~ x'

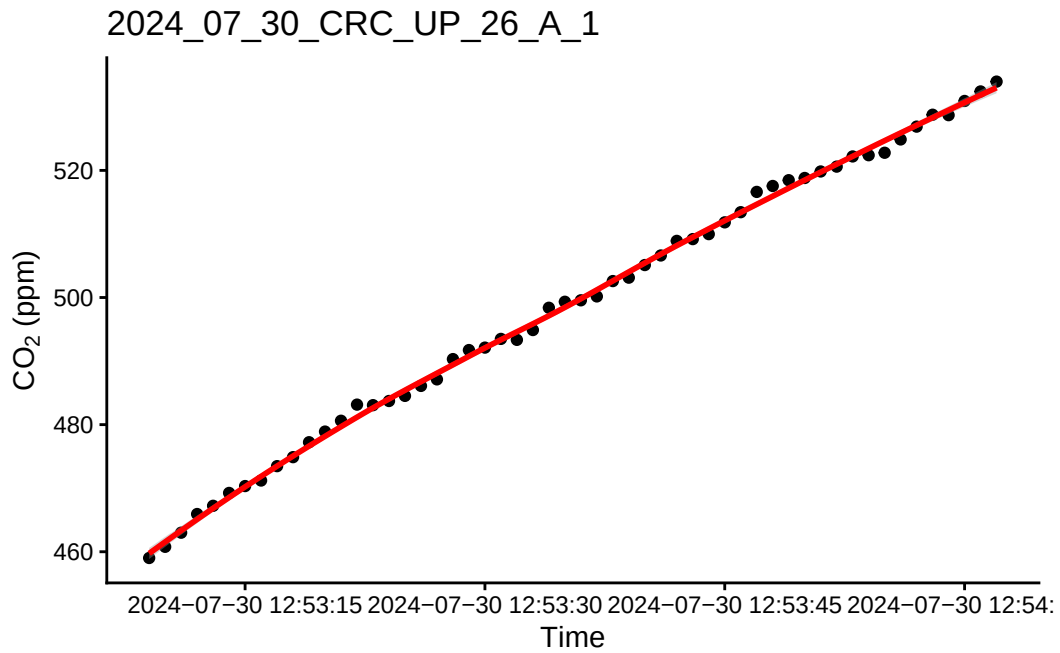


1.2 Check all fluxes

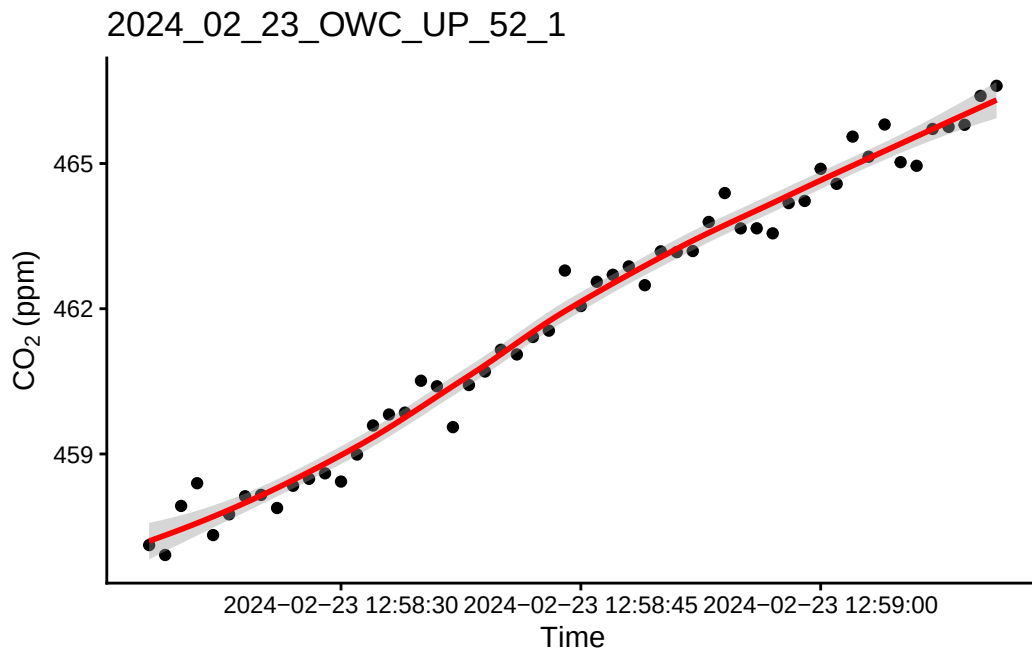
``geom_smooth()`` using method = 'loess' and formula = 'y ~ x'



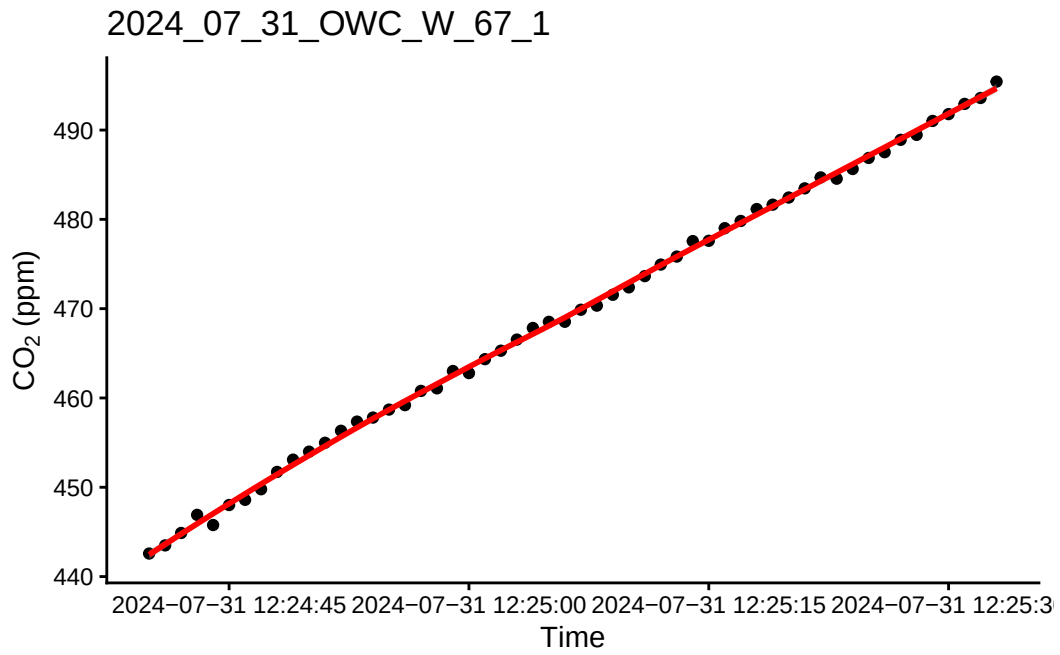
``geom_smooth()`` using method = 'loess' and formula = 'y ~ x'



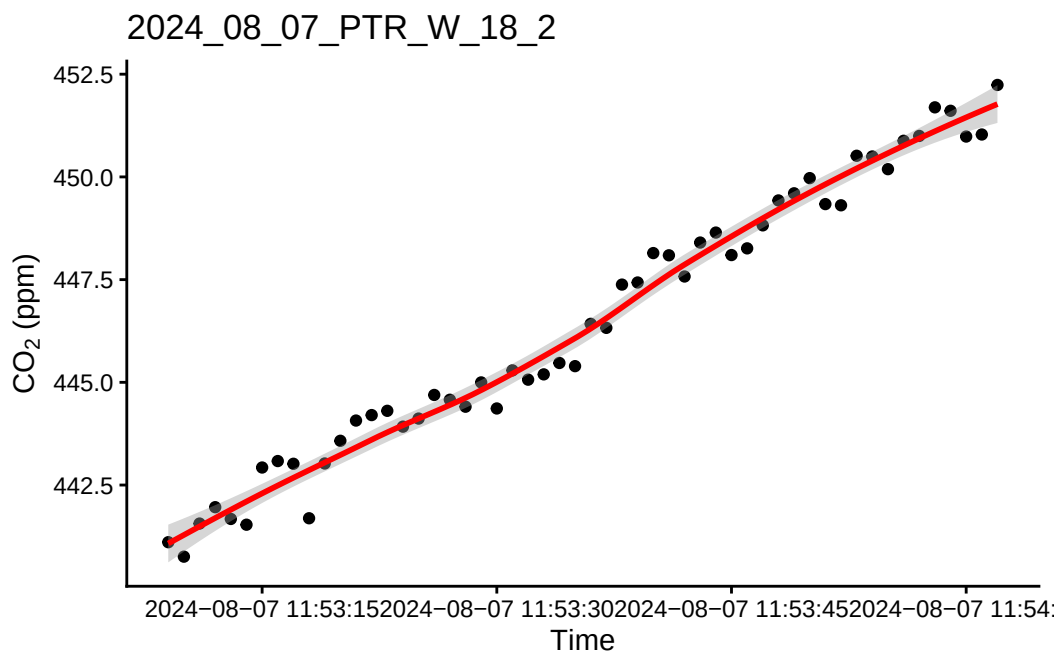
``geom_smooth()`` using method = 'loess' and formula = 'y ~ x'



``geom_smooth()`` using method = 'loess' and formula = 'y ~ x'



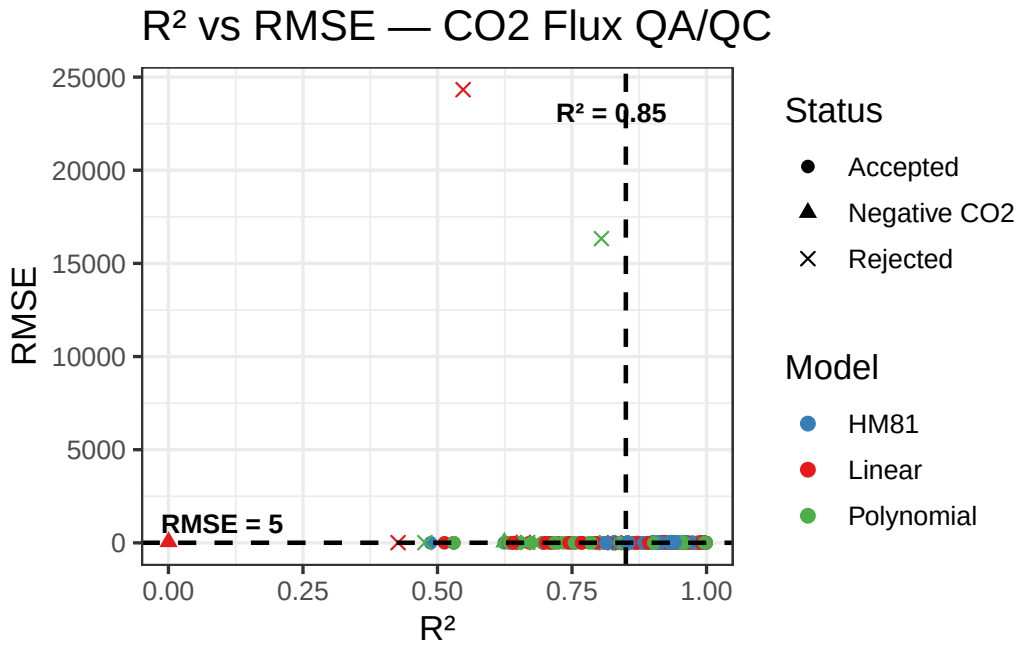
``geom_smooth()`` using method = 'loess' and formula = 'y ~ x'



1.3 QAQC - Flags

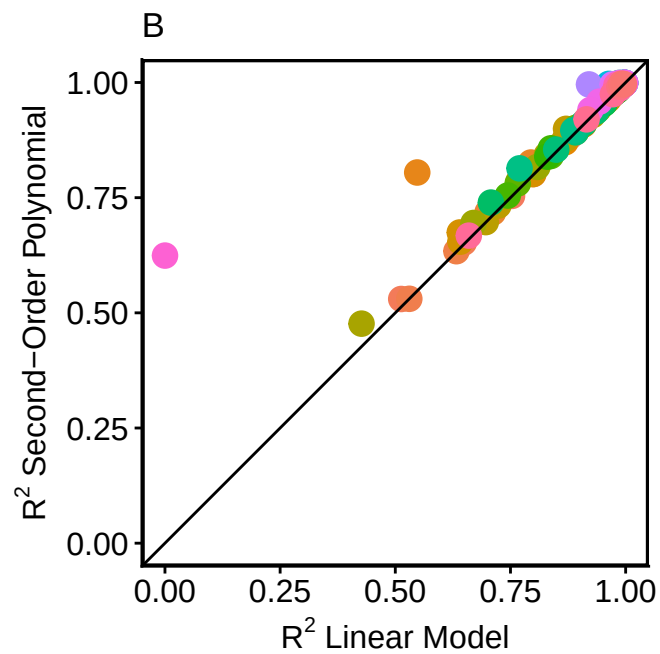
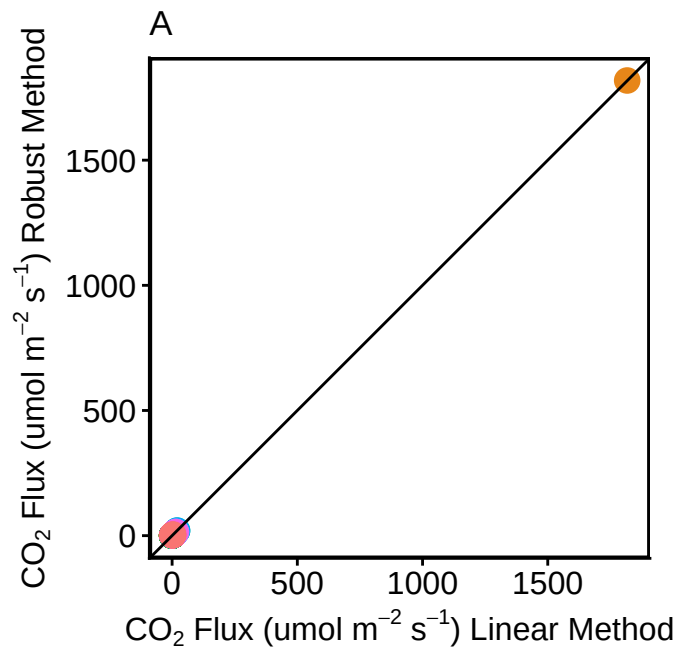
Table 1: QA/QC CO2 Flux Summary

Metric	Value
Total observations	198.0
Rejections	
Rejected — poor model fit (n)	5.0
Rejected — negative CO2 flux (n)	1.0
Rejected — both reasons (n)	1.0
Rejected — total (n)	5.0
Rejected — total (%)	2.5
Rejected - [CO2] (n)	4.0
Rejected — [CO2] (%)	2.0
Model performance	
Passed — Linear model (n)	193.0
Passed — HM81 model (n)	115.0
Passed — Polynomial model (n)	193.0

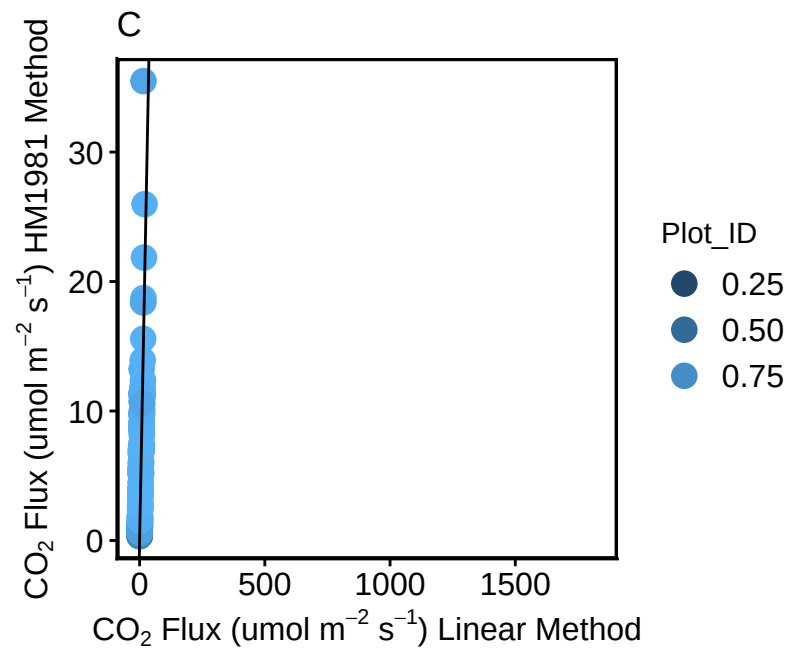


1.4 Adding manual flag to reject any fluxes that do not fit criteria

1.5 Do some visualization of QAQC plots to assess fluxes



Warning: Removed 80 rows containing missing values or values outside the scale range (``geom_point()``).



1.6 Read out final dataframe and export for use elsewhere

```
[1] "TR" "UP" "W"
```

```
[1] 36 37 39 40 48 25 26 27 29 30 41 42 43 47 58 60 61 62 63 NA 50 51 52 55 56
[26] 65 66 67 71 72 33 34 38 28 31 32 44 57 59 64 49 53 54 68 11 12 13 14  2  3
[51]  4  6  7  8 18 19 21 22 23
```

```
[1] 1 2
```